

Omega-3 fatty acids and antioxidants in edible wild plants.



Human beings evolved on a diet that was balanced in the omega-6 and omega-3 polyunsaturated fatty acids (PUFA), and was high in antioxidants. Edible wild plants provide alpha-linolenic acid (ALA) and higher amounts of vitamin E and vitamin C than cultivated plants. In addition to the antioxidant vitamins, edible wild plants are rich in phenols and other compounds that increase their antioxidant capacity. It is therefore important to systematically analyze the total antioxidant capacity of wild plants and promote their commercialization in both developed and developing countries. The diets of Western countries have contained increasingly larger amounts of linoleic acid (LA), which has been promoted for its cholesterol-lowering effect. It is now recognized that dietary LA favors oxidative modification of low density lipoprotein (LDL) cholesterol and increases platelet response to aggregation. In contrast, ALA intake is associated with inhibitory effects on the clotting activity of platelets, on their response to thrombin, and on the regulation of arachidonic acid (AA) metabolism. In clinical studies, ALA contributed to lowering of blood pressure, and a prospective epidemiological study showed that ALA is inversely related to the risk of coronary heart disease in men. Dietary amounts of LA as well as the ratio of LA to ALA appear to be important for the metabolism of ALA to longer-chain omega-3 PUFAs. Relatively large reserves of LA in body fat, as are found in vegans or in the diet of omnivores in Western societies, would tend to slow down the formation of long-chain omega-3 fatty acids from ALA. Therefore, the role of ALA in human nutrition becomes important in terms of long-term dietary intake. One advantage of the consumption of ALA over omega-3 fatty acids from fish is that the problem of insufficient vitamin E intake does not exist with high intake of ALA from plant sources.